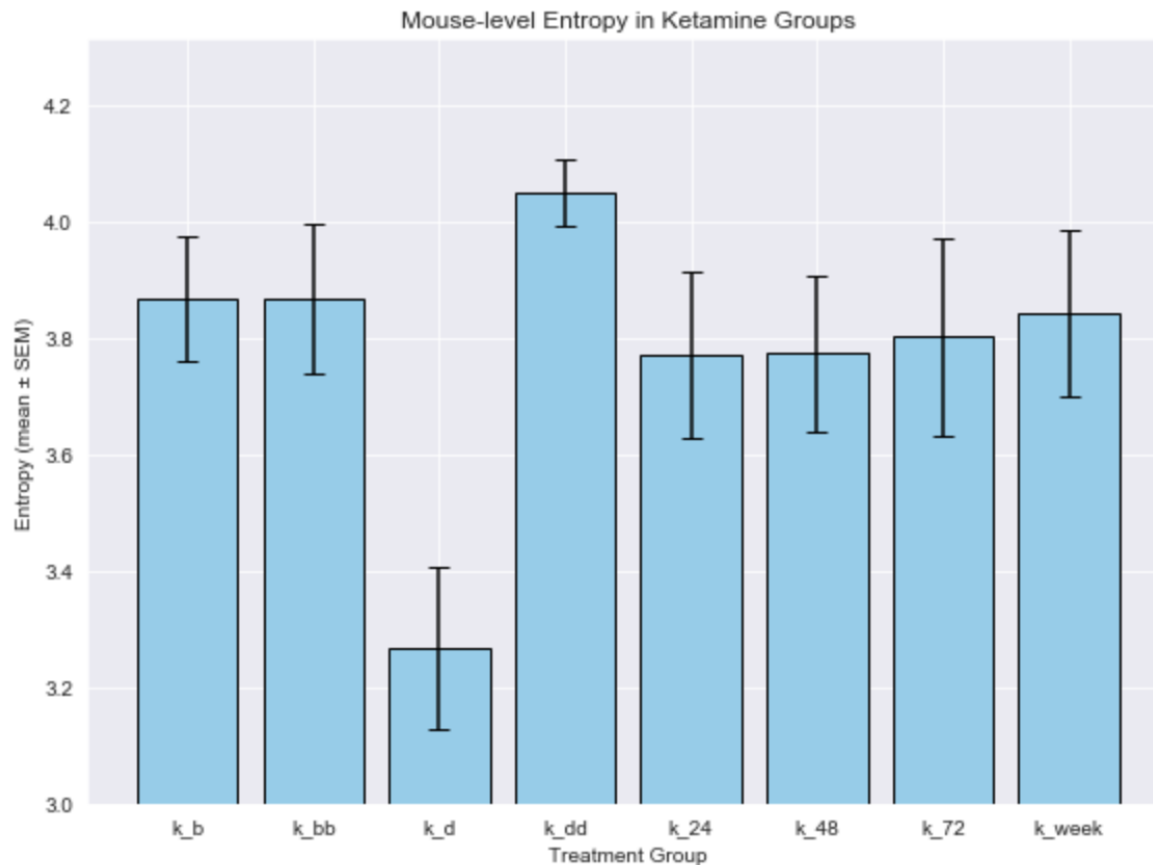


Progress Update

Mohan Yang

Entropy Calculated for each mouse, averaged for the same group



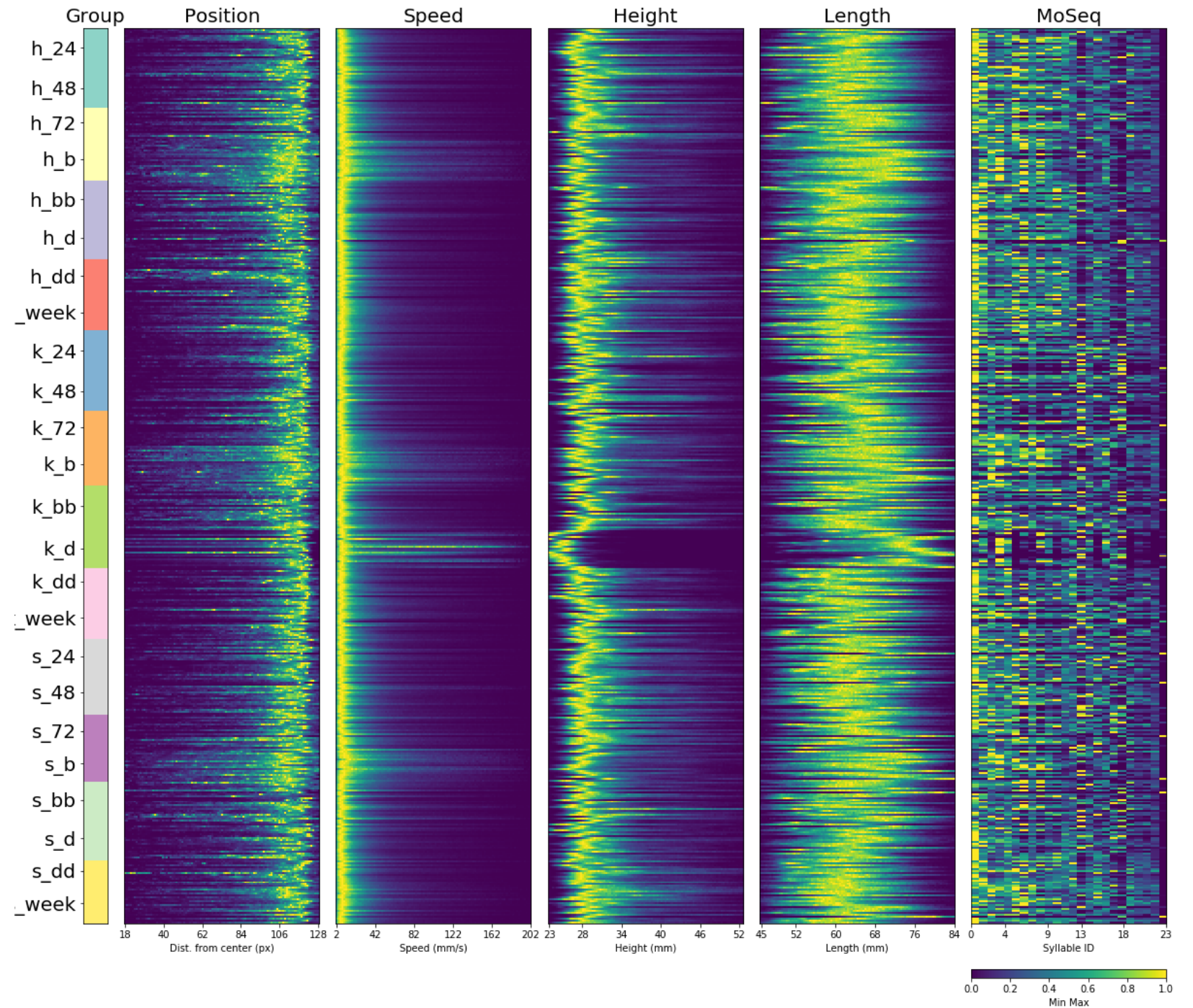
$$\frac{1}{N} \sum_{i=1}^N H(p_i)$$

group	entropy_mean	entropy_sd	n_mice
h_24	4.099375	0.192570	22
h_48	4.083392	0.319621	20
h_72	4.104332	0.316082	17
h_b	3.996420	0.264314	21
h_bb	4.053745	0.336344	20
h_d	3.914212	0.640036	21
h_dd	4.099762	0.227897	19
h_week	4.108343	0.327249	18
k_24	3.770329	0.675177	22
k_48	3.772422	0.601878	20
k_72	3.801084	0.716449	18
k_b	3.866828	0.492771	21
k_bb	3.866774	0.613988	23
k_d	3.266457	0.622381	20
k_dd	4.047720	0.224359	15
k_week	3.841059	0.663399	22
s_24	3.986253	0.626885	18
s_48	4.008672	0.555394	21
s_72	3.943607	0.574606	18
s_b	3.850450	0.581102	17
s_bb	3.840517	0.647131	20
s_d	4.013042	0.290194	21
s_dd	4.193618	0.122436	13
s_week	4.074283	0.539376	20

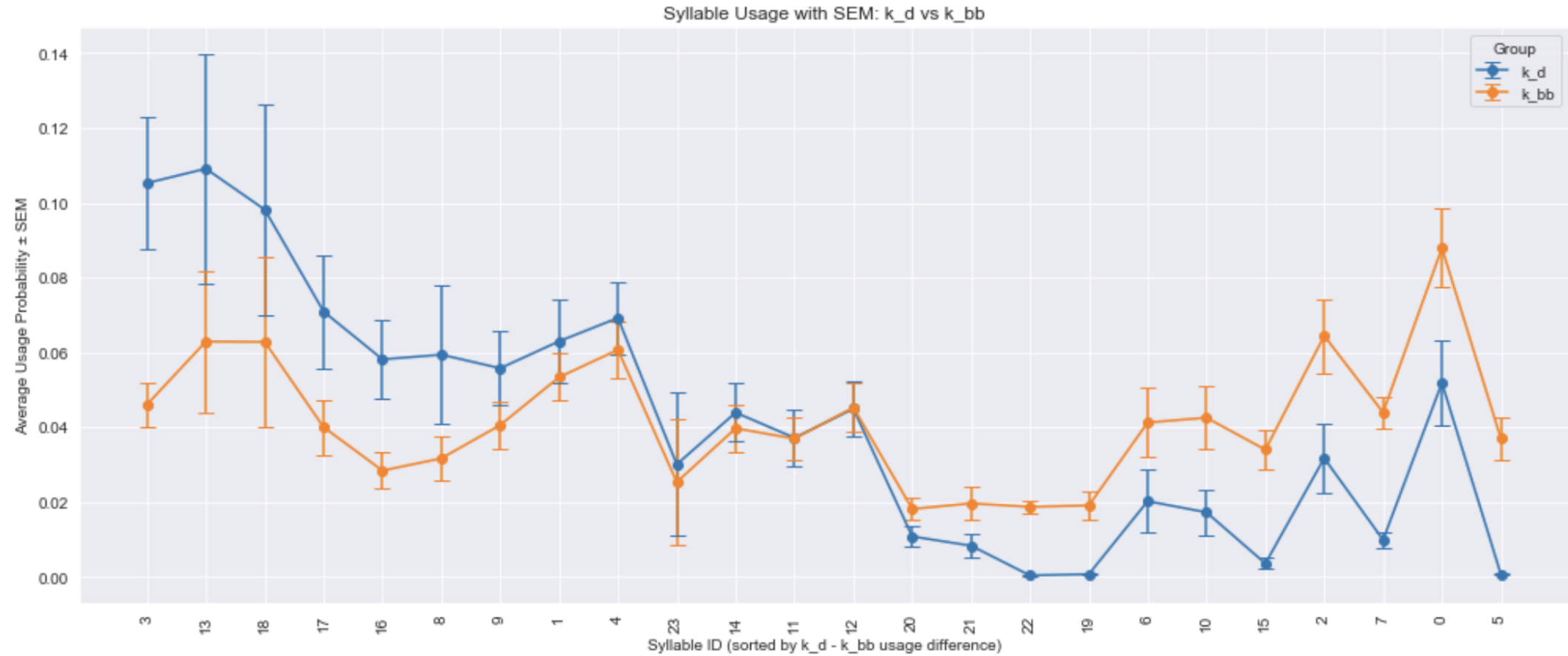
Welch's t-test result: k_d vs k_bb: t = -3.175, p = 0.002885
k_dd vs k_bb: t = 1.288, p = 0.2077

New model's fingerprint plot:

(max_syllable = 25, kappa = 1e+8)



k_bb (before injection) vs. k_d (10 min after injection)

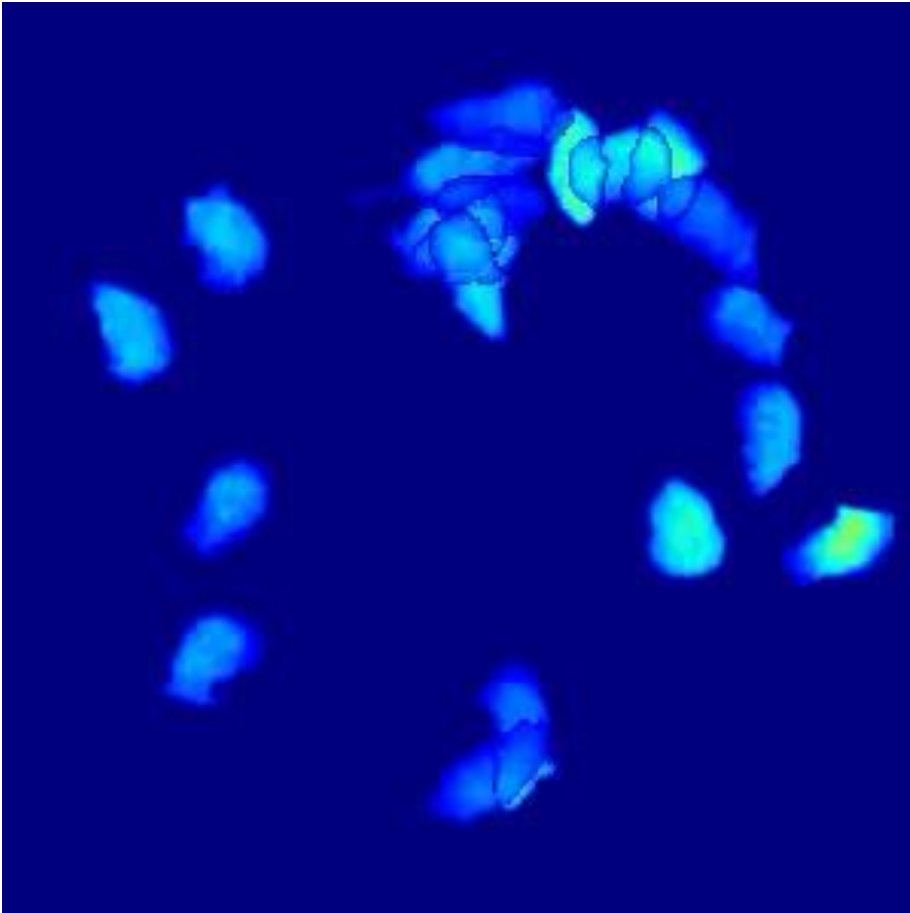


Syllable t-test with FDR adjusted p-value

syllable	t_stat	p_value	fdr_p_value	significant
22	-10.408948	4.174483e-10	1.001876e-08	True
7	-7.481737	1.600150e-08	1.920180e-07	True
5	-6.446250	1.712061e-06	1.369649e-05	True
15	-5.618996	7.343690e-06	4.406214e-05	True
19	-4.984982	5.239402e-05	2.514913e-04	True
3	3.183197	4.114642e-03	1.645857e-02	True
16	2.571939	1.595286e-02	5.469551e-02	False
2	-2.428490	1.963442e-02	5.514371e-02	False
10	-2.412625	2.067889e-02	5.514371e-02	False
0	-2.312002	2.602807e-02	6.246737e-02	False
21	-2.119214	4.069559e-02	8.879038e-02	False
20	-1.833914	7.396543e-02	1.423140e-01	False
17	1.836208	7.708676e-02	1.423140e-01	False
6	-1.690538	9.851698e-02	1.688863e-01	False
8	1.418896	1.692955e-01	2.708728e-01	False
9	1.289460	2.061140e-01	2.944317e-01	False
13	1.283281	2.085558e-01	2.944317e-01	False
18	0.978375	3.341187e-01	4.454916e-01	False
1	0.742357	4.636023e-01	5.856028e-01	False
4	0.692275	4.930278e-01	5.916333e-01	False
14	0.420881	6.762144e-01	7.728164e-01	False
23	0.184924	8.542350e-01	9.318927e-01	False
12	-0.034400	9.727310e-01	9.818558e-01	False
11	0.022901	9.818558e-01	9.818558e-01	False

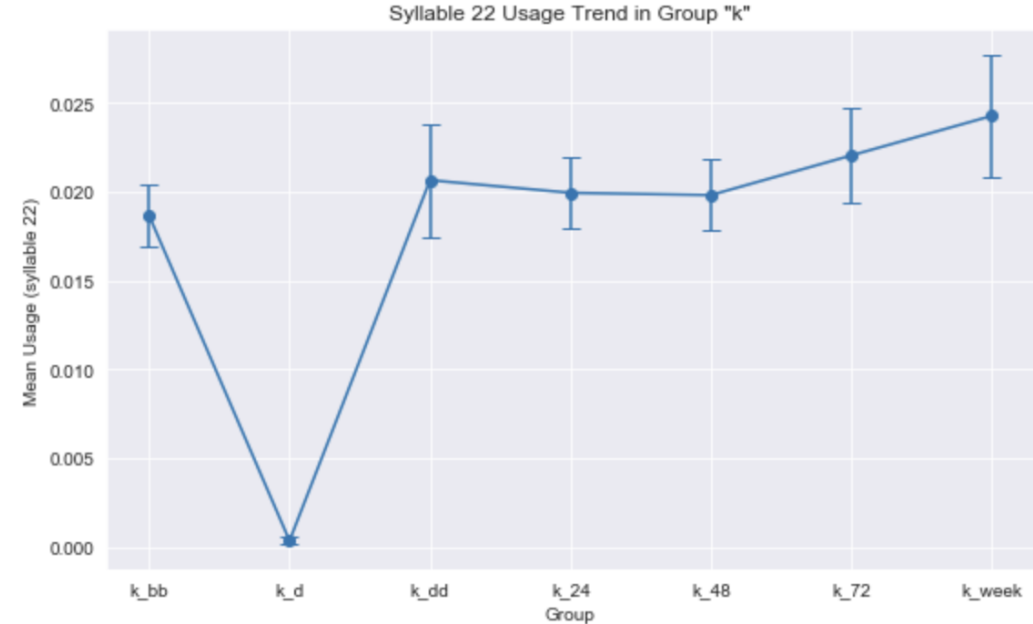
t-stat is computed by $k_d - k_{bb}$

Syllable 22 (grooming)

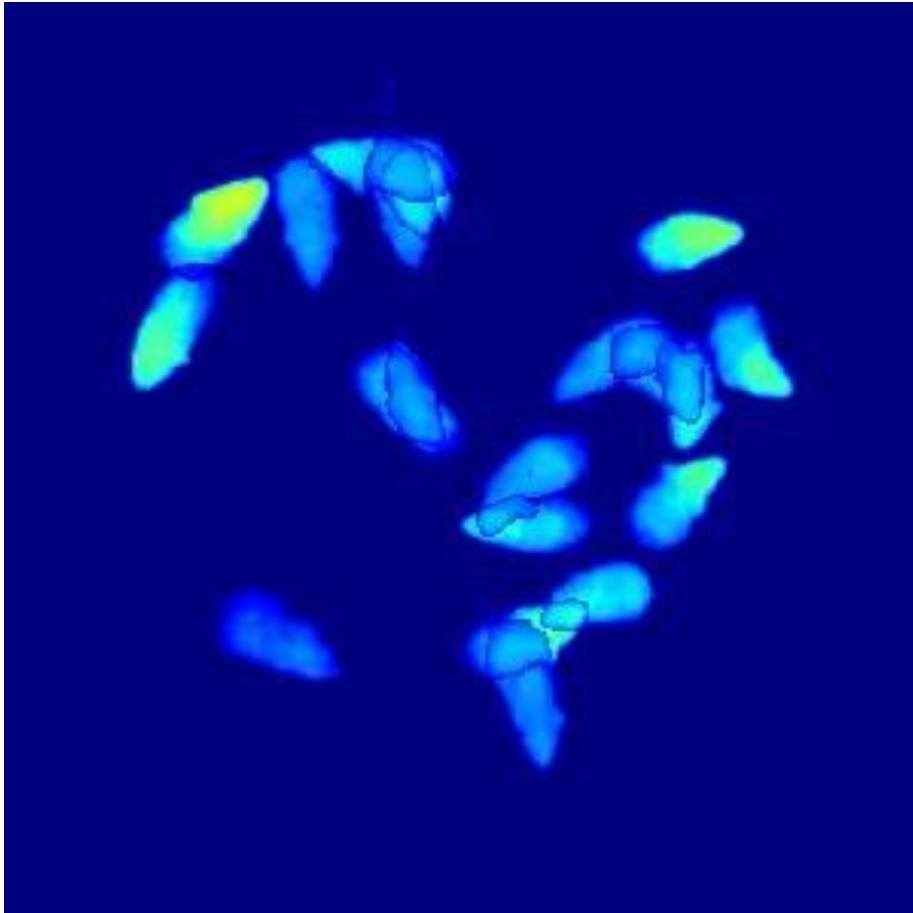


Syllable usage shows grooming behavior is significantly more in k_bb (before injection group) comparing to the k_d (10 min after injection group).

k_dd (2 hour after injection) group has significantly more grooming comparing to k_d group

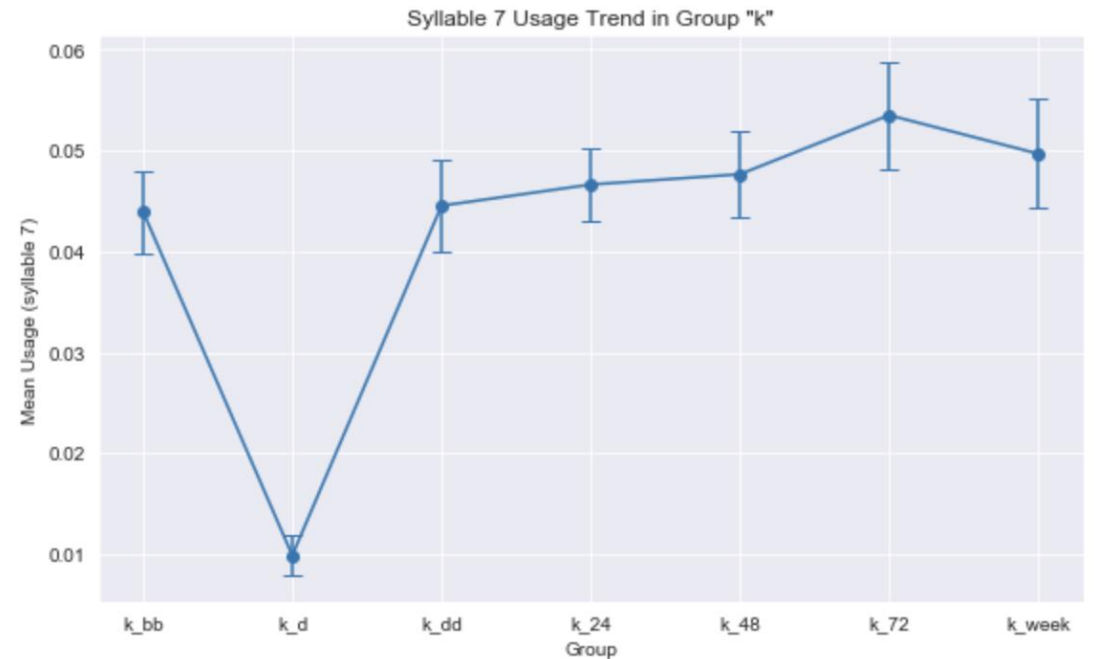


Syllable 7 (rearing)

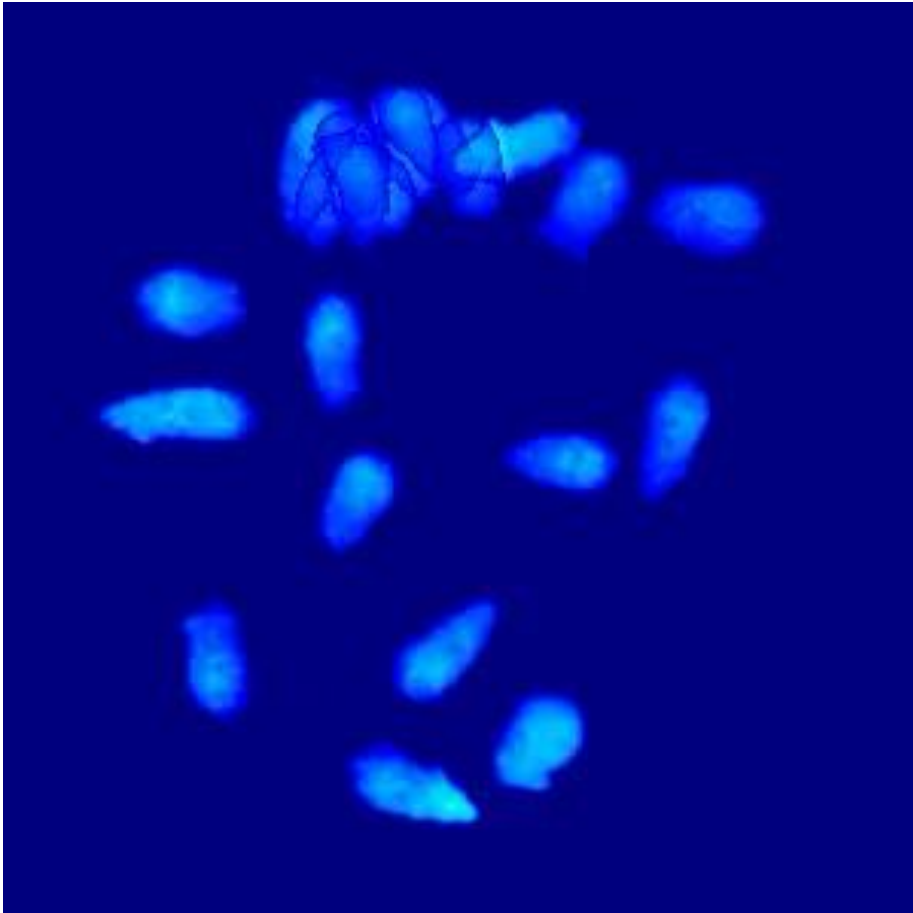


Syllable usage shows rearing behavior is significantly more in k_bb (before injection group) comparing to the k_d (10 min after injection group).

k_dd (2 hour after injection) group has significantly more rearing comparing to k_d group

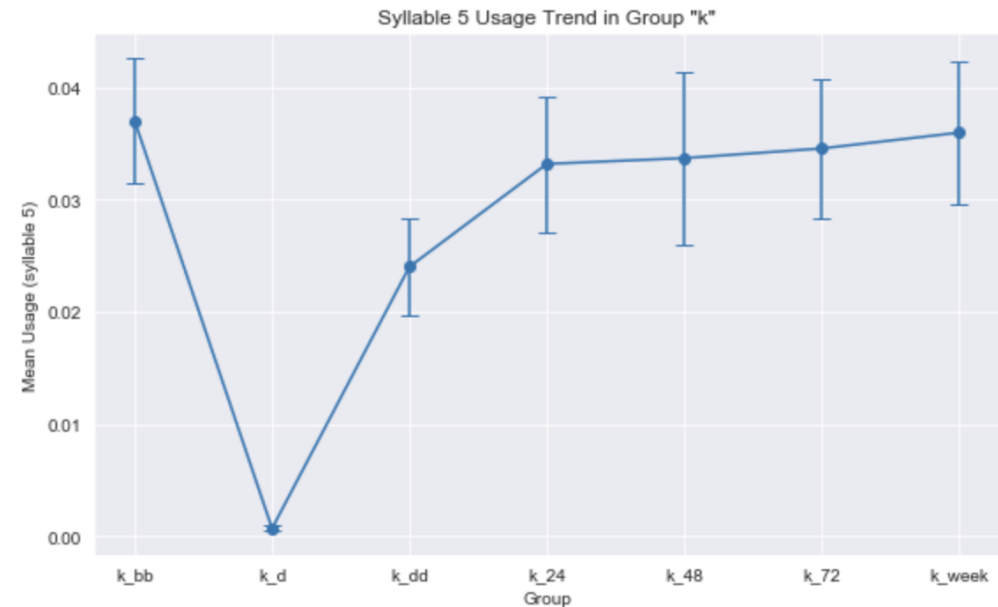


Syllable 5 (staying still)

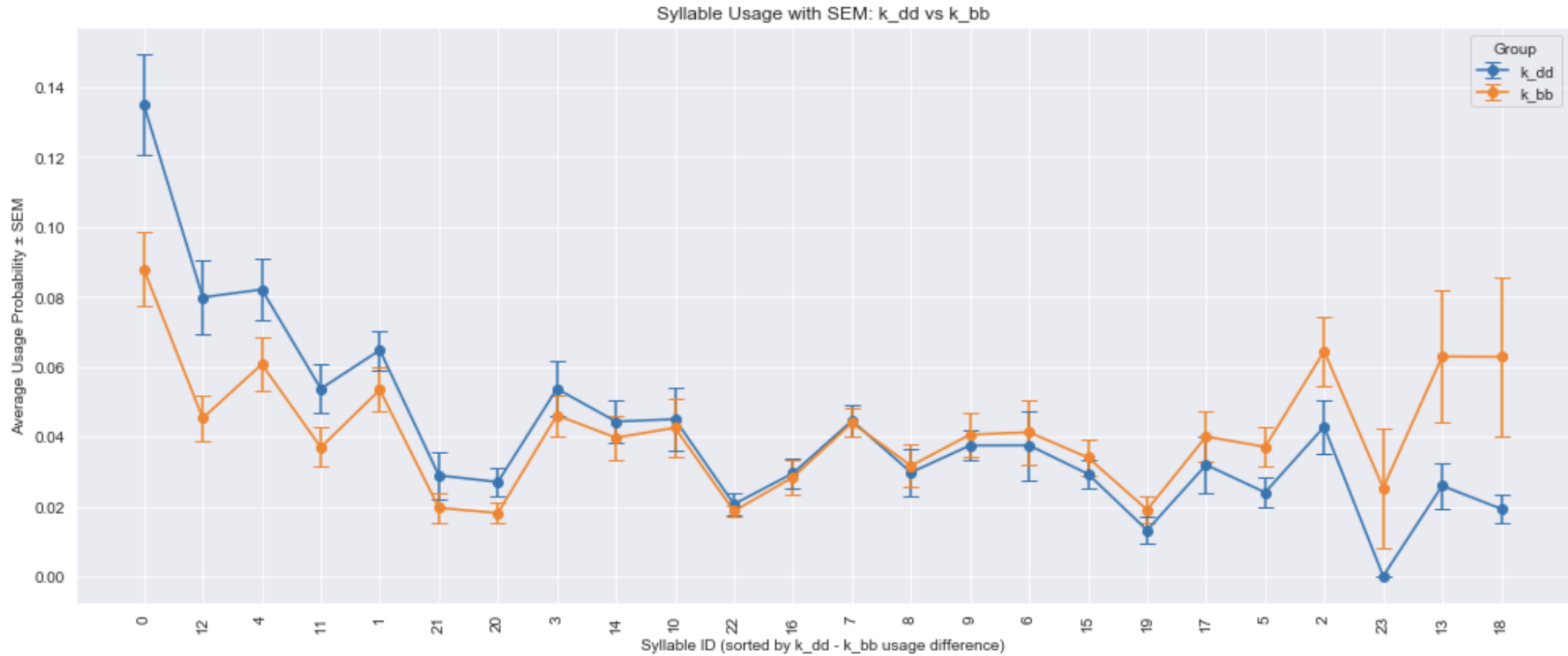


Syllable usage shows staying still behavior is significantly more in k_bb (before injection group) comparing to the k_d (10 min after injection group).

k_dd (2 hour after injection) group has significantly more staying still behavior comparing to k_d group



k_bb (before injection) vs. k_dd (2 hour after injection)

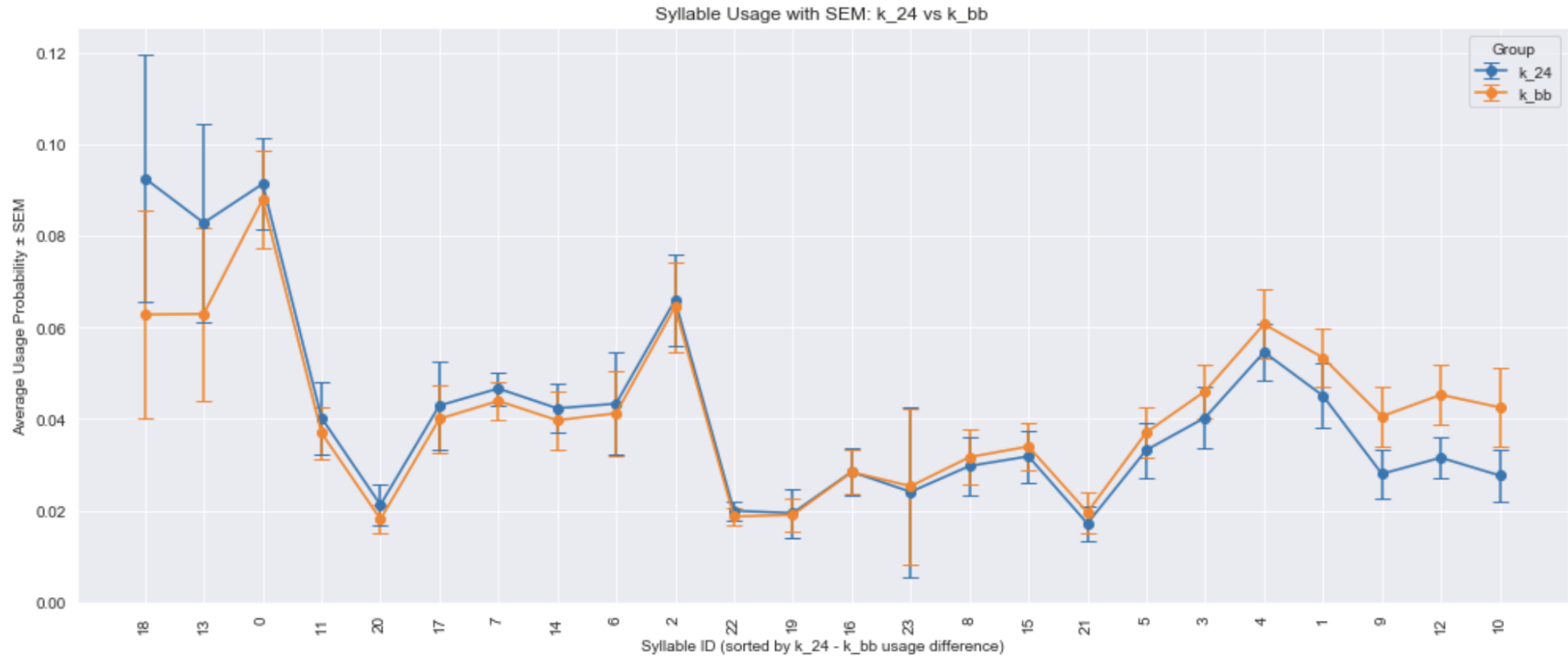


Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
12	12	2.763936	0.010838	0.158091	False
0	0	2.648192	0.013174	0.158091	False
18	18	-1.896595	0.070327	0.257108	False
11	11	1.862646	0.072334	0.257108	False
5	5	-1.839202	0.074162	0.257108	False
13	13	-1.845681	0.075883	0.257108	False
4	4	1.828687	0.076944	0.257108	False
2	2	-1.735686	0.091182	0.257108	False
20	20	1.721023	0.096416	0.257108	False
23	23	-1.485574	0.151582	0.363796	False
1	1	1.347446	0.186339	0.406559	False
21	21	1.142245	0.264031	0.514718	False
19	19	-1.100866	0.278805	0.514718	False
3	3	0.766602	0.449738	0.715888	False
17	17	-0.737098	0.466370	0.715888	False
15	15	-0.718244	0.477259	0.715888	False
22	22	0.542942	0.592548	0.805532	False
14	14	0.523237	0.604149	0.805532	False
9	9	-0.400530	0.691201	0.873096	False
6	6	-0.276628	0.783793	0.899685	False
8	8	-0.212142	0.833320	0.899685	False
10	10	0.194849	0.846709	0.899685	False
16	16	0.174831	0.862198	0.899685	False
7	7	0.096911	0.923403	0.923403	False

t-stat is computed by $k_{dd} - k_{bb}$

k_bb (before injection) vs. k_24 (24 hour after injection)

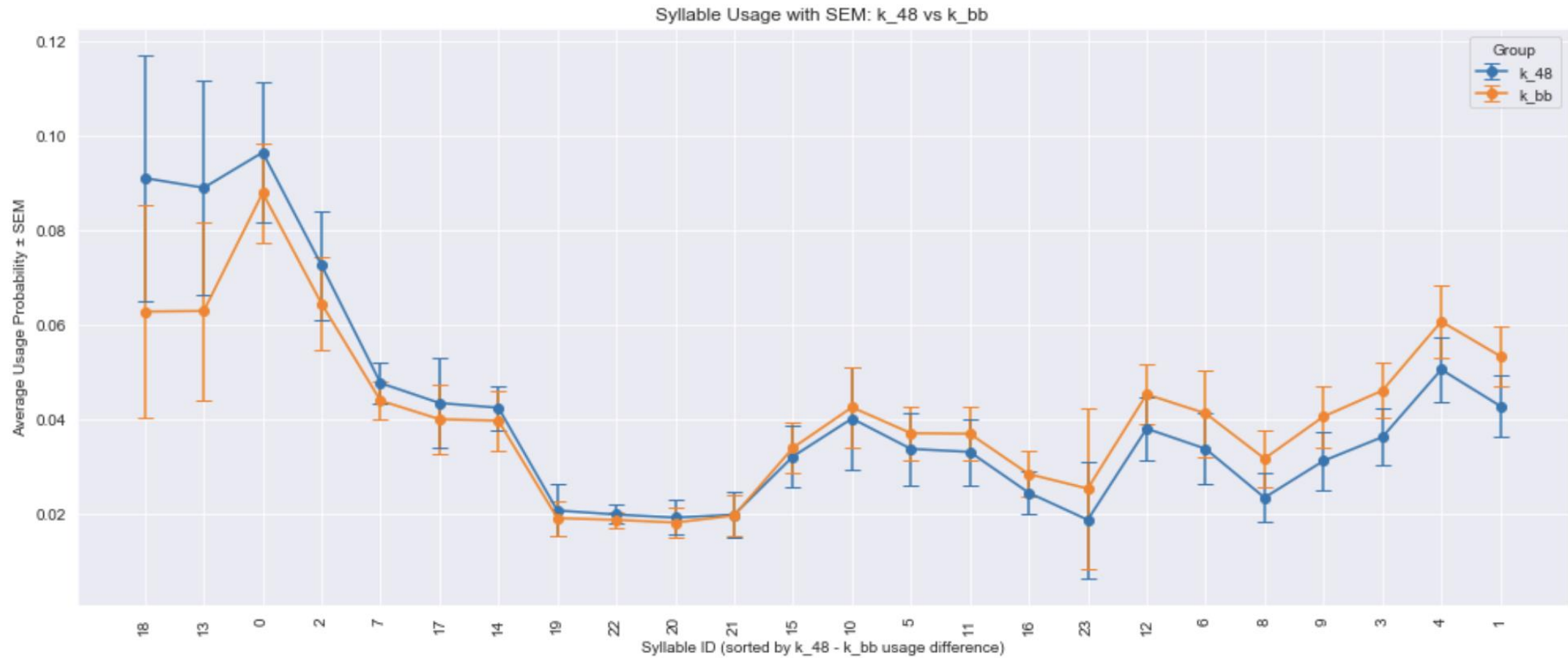


Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
12	12	-1.755866	0.086977	0.993355	False
9	9	-1.509864	0.138601	0.993355	False
10	10	-1.447466	0.155882	0.993355	False
1	1	-0.882221	0.382632	0.993355	False
18	18	0.844379	0.403317	0.993355	False
13	13	0.689717	0.494176	0.993355	False
3	3	-0.653321	0.517101	0.993355	False
4	4	-0.623663	0.536252	0.993355	False
20	20	0.594866	0.555460	0.993355	False
7	7	0.495825	0.622567	0.993355	False
22	22	0.475542	0.636872	0.993355	False
5	5	-0.464061	0.644967	0.993355	False
21	21	-0.423962	0.673758	0.993355	False
11	11	0.337834	0.737331	0.993355	False
14	14	0.317523	0.752415	0.993355	False
15	15	-0.277537	0.782715	0.993355	False
17	17	0.240174	0.811431	0.993355	False
0	0	0.234444	0.815754	0.993355	False
8	8	-0.218237	0.828283	0.993355	False
6	6	0.144365	0.885920	0.993355	False
2	2	0.107579	0.914830	0.993355	False
19	19	0.061283	0.951457	0.993355	False
23	23	-0.048827	0.961287	0.993355	False
16	16	0.008377	0.993355	0.993355	False

t-stat is computed by $k_{24} - k_{bb}$

k_bb (before injection) vs. k_48 (48 hour after injection)

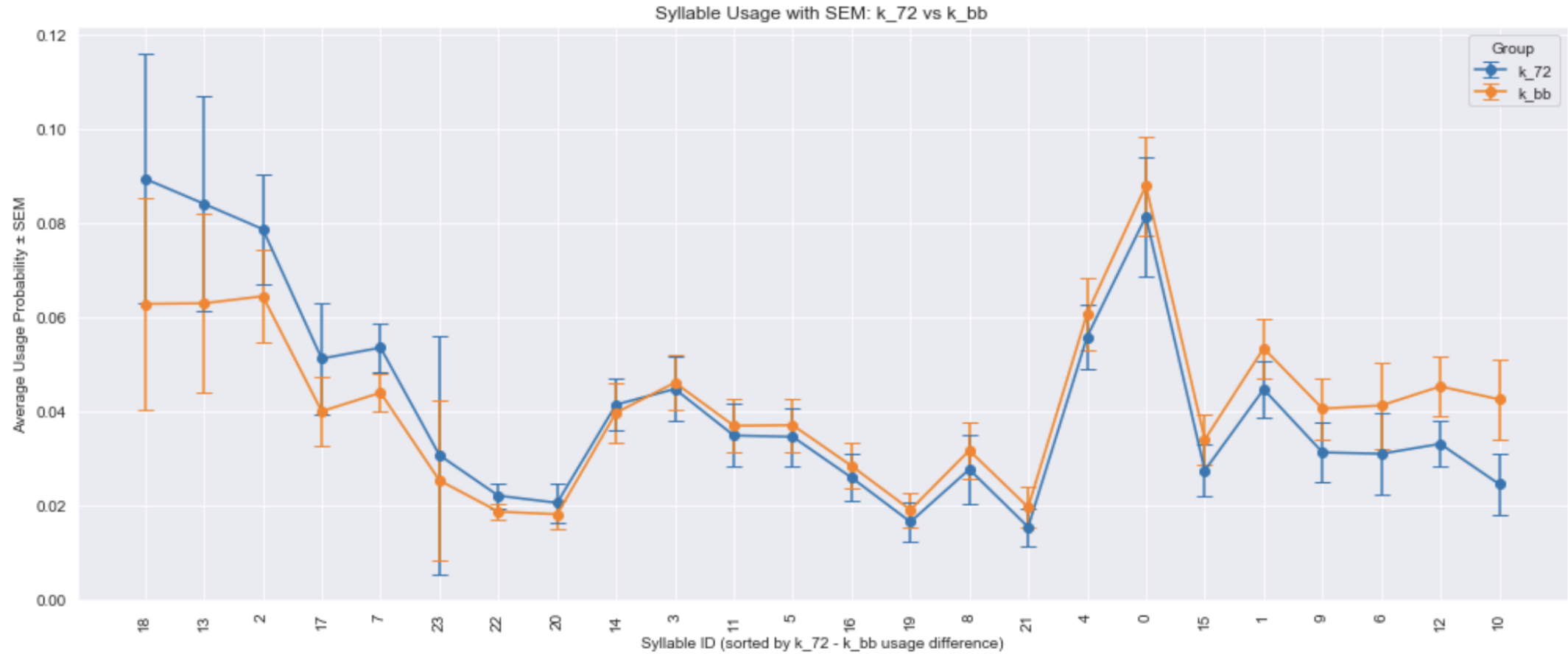


Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
3	3	-1.176070	0.246407	0.900832	False
1	1	-1.166662	0.250151	0.900832	False
9	9	-1.054348	0.297903	0.900832	False
8	8	-1.031921	0.308201	0.900832	False
4	4	-0.990336	0.327824	0.900832	False
13	13	0.881232	0.383671	0.900832	False
18	18	0.821462	0.416348	0.900832	False
12	12	-0.776494	0.441988	0.900832	False
7	7	0.633605	0.529899	0.900832	False
6	6	-0.631688	0.531150	0.900832	False
16	16	-0.599546	0.552108	0.900832	False
2	2	0.535807	0.595135	0.900832	False
0	0	0.471787	0.639997	0.900832	False
22	22	0.431697	0.668315	0.900832	False
11	11	-0.430091	0.669556	0.900832	False
14	14	0.348282	0.729497	0.900832	False
5	5	-0.348341	0.729622	0.900832	False
23	23	-0.314351	0.754935	0.900832	False
17	17	0.284118	0.777909	0.900832	False
19	19	0.244601	0.808236	0.900832	False
15	15	-0.222644	0.825003	0.900832	False
20	20	0.221653	0.825762	0.900832	False
10	10	-0.172703	0.863815	0.901373	False
21	21	0.026922	0.978657	0.978657	False

t-stat is computed by $k_{48} - k_{bb}$

k_bb (before injection) vs. k_72 (72 hour after injection)



Syllable t-test with FDR adjusted p-value

	syllable	t_stat	p_value	fdr_p_value	significant
10	10	-1.686428	0.099862	0.881592	False
12	12	-1.512485	0.138626	0.881592	False
7	7	1.433953	0.160743	0.881592	False
22	22	1.059881	0.297546	0.881592	False
9	9	-1.034207	0.307484	0.881592	False
1	1	-1.001029	0.323012	0.881592	False
2	2	0.935044	0.355999	0.881592	False
15	15	-0.870496	0.389495	0.881592	False
6	6	-0.816525	0.419179	0.881592	False
17	17	0.803634	0.428070	0.881592	False
18	18	0.762699	0.450618	0.881592	False
13	13	0.713230	0.480373	0.881592	False
21	21	-0.709376	0.482314	0.881592	False
4	4	-0.481582	0.632795	0.881592	False
20	20	0.478031	0.635770	0.881592	False
19	19	-0.465015	0.644656	0.881592	False
8	8	-0.424134	0.674007	0.881592	False
0	0	-0.398485	0.692659	0.881592	False
16	16	-0.352752	0.726233	0.881592	False
5	5	-0.295210	0.769475	0.881592	False
11	11	-0.236892	0.814089	0.881592	False
14	14	0.204362	0.839133	0.881592	False
23	23	0.175723	0.861658	0.881592	False
3	3	-0.150002	0.881592	0.881592	False

t-stat is computed by $k_{72} - k_{bb}$